

EDUCATION	University of Illinois at Urbana-Champaign <i>Ph.D. in Economics</i> Advisor: Prof. Tatyana Deryugina	Champaign, IL May 2026 (<i>expected</i>)
	University of Illinois at Urbana-Champaign <i>M.S. in Policy Economics</i>	Champaign, IL Dec 2019
	University of International Business and Economics (UIBE) <i>B.S. in Economics (Minor in French)</i>	Beijing, China Jun 2018
	University of California, Berkeley <i>Exchange student</i>	Berkeley, CA Jan 2017 - May 2017

RESEARCH INTERESTS **Environmental Economics, Public Economics, Applied Econometrics**

WORKING PAPERS **Air Pollution and Economic Activity: Evidence from Foot Traffic Patterns in the U.S.** (Job Market Paper)

I investigate how air pollution affects economic activity. Using over 600 million phone-location-based foot traffic data points from SafeGraph, I conduct a large-scale analysis to examine the causal effect of air pollution on activity patterns across the US. Using changes in local wind direction as an instrumental variable (IV) for air pollution, I characterize the dynamic response to pollution exposure. I find that PM_{2.5} reduces economic activity in both the short and medium run, with effects persisting for up to two weeks before gradually recovering. The reductions are widespread across different economic sectors, with recreational activities experiencing the largest decline. The effect is more pronounced in higher-income counties and areas with larger shares of children, suggesting greater awareness among wealthier or more vulnerable populations.

Go with the Wind: Polluters' Strategic Response to Wind Directions

I investigate the strategic behavior of polluters in the US. Under the Clean Air Act, air quality monitor data determine whether an area meets environmental standards. Because wind carries pollution, monitors detect more pollution from upwind sources and less from downwind sources. This disparity incentivizes polluters to emit less when they are upwind of a monitor, and more when they are downwind. I identify such strategic behavior among US coal-fired power plants, finding that a one-standard-deviation increase in favorable wind direction—when wind blows pollutants away from monitors—leads to a 0.8% (172 lbs) and 0.4% (54 lbs) increase in sulfur dioxide (SO_2) and nitrogen dioxide (NO_x) emissions, respectively. At the same time, fuel input remains constant while emission rates rise, suggesting that power plants temporarily turn off pollution control equipment. Additionally, the increase is more pronounced when power plants are in a non-attainment county, located in the same state as the monitor, or surrounded by fewer nearby polluters. These findings suggest that polluters facing stricter regulatory pressure are more likely to respond strategically when conditions are favorable.

WORK IN PROGRESS

Natural Disasters and Occupational Mobility: Evidence from the 1927 Mississippi Flood (joint with Xiaohan Wang)

We investigate how a major natural disaster affect intergenerational occupational mobility in the agricultural sector. Linking individuals across the 1920, 1930, and 1940 Censuses, we show that children residing in counties affected by the 1927 Mississippi Flood were significantly less likely to become farmers as adults. This effect is mainly driven by children from non-farming families, who were 9% less likely to work in agriculture and 10% more likely to transition into the service sector. In the longer run, these individuals achieved socioeconomic status scores about 3% higher than comparable children in non-flooded counties. Our findings suggest that natural disasters can accelerate structural change by shifting new labor market entrants out of agriculture and into higher-status occupations.

Climate Change and Agricultural Adaptation in Peru

(joint with Tatyana Deryugina)

Social Networks and Household Protective Investments: Evidence from Wildfire Smoke

(joint with Sanjukta Mitra)

Deforestation and Air Pollution in the Indonesian Rainforest

(joint with Francesco Cenerini)

CONFERENCES AND WORKSHOPS

2026

ASSA Annual Meeting (Philadelphia)

2025

Canadian Resource and Environmental Economics Association Annual Conference (Winnipeg, Canada), Southern Economic Association Annual Conference (Tampa), Heartland Workshop (poster, UIUC), Program in Environmental and Resource Economics Seminar (UIUC), Midwest Economics Association Annual Conference (Kansas City), Eastern Economic Association Annual Conference (New York City), Applied Micro Research Lunch (UIUC), Graduate Student Research Seminar (UIUC)

2024

Occasional Workshop in Environmental and Resource Economics (UCSB), Online Summer Workshop in Environment, Energy, and Transportation (Virtual), Applied Micro Research Lunch (UIUC), Graduate Student Research Seminar (UIUC)

2023

Price Theory Summer Camp (University of Chicago), Applied Micro Research Lunch (UIUC)

2022

Berkeley/Solan Summer School in Environmental and Energy Economics, Applied Micro Research Lunch (UIUC)

TEACHING EXPERIENCE

ECON 102: Microeconomic Principles

Teaching Assistant

Fall 2021, Spring 2022

The Economics of the Firm (EMBA at the University of Warsaw)

Teaching Assistant to Prof. Hadi Esfahani

Fall 2019

Econ 528: Microeconomics for Business

Course Assistant

Spring 2019, Summer 2019

WORKING EXPERIENCE	Gies College of Business, UIUC Research Assistant to Prof. Tatyana Deryugina	Aug 2022 – present
	Department of Economics, UIUC Research Assistant to Prof. Eunyi Chung	June 2021 – Aug 2022
	China Institute for Educational Finance Research, Peking University Research Associate	Mar 2020 – Aug 2020
	Risk Management Department, Bank of Communications Data Analyst Intern	Jun 2017 – Aug 2017
	School of International Trade and Economics, UIBE Research Assistant to Prof. Deyu Yuan	Sep 2016 – Jan 2017
AWARDS	• Robert Willis Harbeson Memorial Dissertation Fellowship, University of Illinois 2025 • Hung-Chao and Julia Tai Family Fellowship, University of Illinois 2025 • Cleo Fitzsimmons Award, University of Illinois 2022 • Graduate Fellowship, University of Illinois 2020	
SKILLS	Languages: Chinese (Native), English (Proficient), French (Basic). Programming: R, Python, Stata, Git, Shell.	
REFERENCES	Tatyana Deryugina (Chair) Associate Professor of Finance Gies College of Business University of Illinois at Urbana-Champaign George Deltas Professor Department of Economics University of Illinois at Urbana-Champaign David Molitor Associate Professor of Finance Gies College of Business University of Illinois at Urbana-Champaign Julian Reif Associate Professor of Finance Gies College of Business University of Illinois at Urbana-Champaign	

Last Updated: Sep 15, 2025